

## INDIAN SPACE RESEARCH ORGANISATION (ISRO) — SIH26169

# HORIZON User Manual & Evaluator Test Execution Guide

Step-by-step instructions for running, testing, and evaluating the HORIZON Tracking Suite

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## 1. System Requirements & Launch Instructions

### A. Standalone Executable (No Python Required):

Double-click dist\HORIZON\HORIZON.exe. The system launches immediately offline without cloud or internet connection.

### B. Python Source Launch:

Open terminal in coarse-align-x directory and run: `python main.py`.

## 2. Workstation Navigation (Bank Switcher)

HORIZON is organized into 5 specialized workstations accessible from the top bank switcher:

- **Mode 1 (Mission Setup):** Select from 13 predefined space/airborne scenarios, customize trajectory patterns, and set beacon geometry.
- **Mode 2 (Live Tracking):** Core dual-feed tracking workstation displaying clean ground-truth vs. Phase 4 perception feed.
- **Mode 3 (Track Geometry):** Kalman state covariance ellipses, innovation gates, and coordinate transforms.
- **Mode 4 (Scientific Benchmarks):** Live 4-way algorithmic comparison suite and PDF report exporter.
- **Mode 5 (Stress Testing):** Manual adversarial parameter injection sliders.

## 3. Evaluating External MP4 Test Videos (ISRO Evaluation-2)

To test your own test video on a USB drive:

1. Navigate to **Live Tracking (Mode 2)**.
2. In the right sidebar, locate **'Video Input Source (ISRO Evaluation-2)'**.
3. Select **'EXTERNAL\_VIDEO'** from the dropdown.
4. Click **'Load External MP4...'** and select your .mp4 video file (sample available at `data/samples/isro_sample_beacon_test.mp4`).
5. Click **'Resume'**. The system processes the video frame-by-frame in real-time, detecting the optical beacon centroid, updating the Kalman filter, and displaying tracking overlays.
6. Click **'Export Centroid Log (CSV)'** to save the frame-by-frame centroid coordinates, errors, and FPS to a CSV file.

## 4. Exporting Official ISRO PDF Performance Reports

To generate formal documentation for judges:

1. Switch to **Scientific Benchmarks (Mode 4)**.
2. Click **'Run Live 4-Way Benchmark Suite'** to execute live empirical evaluation across all 4 algorithms.
3. Click **'Export ISRO PDF Performance Report'** to save `HORIZON_ISRO_Performance_Report.pdf`.
4. Choose **'Open'** when prompted to immediately view the generated publication-quality document.

## 5. Troubleshooting & Diagnostics

- **Video Codec:** HORIZON supports MP4 (H.264 / MPEG-4), AVI, MOV, and MKV. If a video fails to open, confirm standard OpenCV-readable encoding.

- **Detection Blackout Test:** Press '**■ Suppress Detection**' on the Live screen to verify the Kalman filter holds boresight lock during simulated optical signal loss.